

4.7.5 Food production (biology only)

4.7.5.1 Factors affecting food security

Content	Key opportunities for skills development
Students should be able to describe some of the biological factors affecting levels of food security. Food security is having enough food to feed a population. Biological factors which are threatening food security include: • the increasing birth rate has threatened food security in some countries • changing diets in developed countries means scarce food resources are transported around the world • new pests and pathogens that affect farming • environmental changes that affect food production, such as widespread famine occurring in some countries if rains fail • the cost of agricultural inputs • conflicts that have arisen in some parts of the world which affect the availability of water or food. Sustainable methods must be found to feed all people on Earth.	WS 1.4 Interpret population and food production statistics to evaluate food security.

4.7.5.2 Farming techniques

Content	Key opportunities for skills development
The efficiency of food production can be improved by restricting energy transfer from food animals to the environment. This can be done by limiting their movement and by controlling the temperature of their surroundings. Some animals are fed high protein foods to increase growth.	WS 1.3 Understand that some people have ethical objections to some modern intensive farming methods. WS 1.4 Evaluate the advantages and disadvantages of modern farming techniques.

4.7.5.3 Sustainable fisheries

Content	Key opportunities for skills development
Fish stocks in the oceans are declining. It is important to maintain fish stocks at a level where breeding continues or certain species may disappear altogether in some areas. Control of net size and the introduction of fishing quotas play important roles in conservation of fish stocks at a sustainable level.	WS 1.4 Understand how application of different fishing techniques promotes recovery of fish stocks.

4.7.5.4 Role of biotechnology

Content	Key opportunities for skills development
Students should be able to describe and explain some possible biotechnical and agricultural solutions, including genetic modification, to the demands of the growing human population. Modern biotechnology techniques enable large quantities of microorganisms to be cultured for food. The fungus <i>Fusarium</i> is useful for producing mycoprotein, a protein-rich food suitable for vegetarians. The fungus is grown on glucose syrup, in aerobic conditions, and the biomass is harvested and purified. A genetically modified bacterium produces human insulin. When harvested and purified this is used to treat people with diabetes. GM crops could provide more food or food with an improved nutritional value such as golden rice.	There are links with this content to Genetic engineering.